PeerForge — Pitch Deck

Speaker notes in italics. Every figure is measured against a running instance, 2026-08-05.

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1 · Title

PeerForge

Face the panel before the panel faces you

An AI peer-review panel that argues about your research — and checks its own work.

Open with the demo, not the slide. Upload a paper, run three turns, let them watch reviewers disagree.

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2 · The problem

Serious criticism arrives too late to act on.

- A PhD student waits weeks for supervisor comments, months for reviewer

reports.

- By then the design is fixed, the data collected, the flaw unfixable.
- The single most useful sentence — *"you cannot claim causation with this

allocation procedure"* — lands after it is any use.

Supervisors are stretched. Examiners see the work once. The gap between **"I think it's ready"** and **"the panel thinks it's ready"** is where viva failures and desk rejections live.

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3 · Why the obvious fix fails

Ask a general AI to review your paper and you get:

> "This is a well-structured study. Consider expanding the literature review. As noted in Section 2.3 on page 12, the sampling approach..."

The paper has no Section 2.3. It has no page 12.

This is the slide that lands. Everyone in the room has seen this output.

Point five models at one paper and it gets worse: within two rounds you have one opinion restated five times, everyone agreeing, and confident citations to nothing.

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4 · What we built

A panel, not an assistant.

Each reviewer has a remit and stays in it:

- **Methodologist** — allocation, power, controls, validity threats
- **Domain expert** — construct validity, related work, novelty
- **External examiner** — is this publishable, and what must change

They read your document, research the literature, **prepare privately**, then disagree with each other in front of you.

Then the part nobody else does: **it checks itself.**

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5 · The hard part

Eleven integrity rules, every turn.

Repetition · deference · self-contradiction · role drift · generic filler · fabricated citations · contradicted citations · session meta-commentary

Ten force regeneration. The retry is **re-checked** — a retry that fixes one problem and introduces another is still a failure.

Citations are verified against the document.

Situation	What happens
No document uploaded	Every page reference is fabrication → stripped
Document uploaded	Cited sections checked against what it **actually contains**
Section 9 of a 4-section paper	Caught
Section 2.1 that genuinely exists	Passes untouched
`(McKay et al., 2018)`	Never flagged — a reviewer may know it

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6 · The proof

This is the slide to slow down on.

	Before	After
Fabricated citations reaching the transcript	**33** across 18 turns	**0**
Contradicted citations	**11**	**0**
Panel restatement rate	**1.00**	**0.20**
Reviewers opening by deferring	**0.80**	**0.00**
Retrieved sources actually cited	**0 of 5**	**3 of 5**

The fabrication test was **adversarial**: three reviewers explicitly instructed to cite pages, sections, tables and figures in every paragraph, on a session with no document at all. Zero got through.

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7 · What the user gets

1 **Transcript** — the argument as it happened
2 **Summary** — grounded in the real transcript; refuses to summarise a

session that did not happen

1 **15 defence questions** — each tied to the excerpt from **your** document

that prompted it

1 **Readiness assessment** — ten scored dimensions with a stated basis
2 **Signed certificate** — publicly verifiable, hash-bound to its evidence

Show a real certificate verifying. Then change the evidence and show it report `SUPERSEDED` **— authentic but stale, not forged.**

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8 · Market

> ⚠ **Sizing figures below are placeholders — source them before you present this.** Every other number in this deck was measured against a running instance; these were not, and an investor will check them.

Primary — doctoral candidates. A viva is a single high-stakes event with almost no rehearsal available. **[Insert enrolment figures for your target geographies.]**

Secondary — researchers pre-submission. Desk rejection is common at selective journals, and much of it is for reasons a panel would name in ten minutes. **[Insert desk-rejection rates for your target venues.]**

Institutional — departments and graduate schools. Courses, roles, seats and invitations are built in. A department buys seats; professors run courses; students see their own sessions.

Land-and-expand: one supervisor's cohort, then the department, then the graduate school.

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9 · Business model

Institutional seats. Priced per billable student seat; staff seats are free. Seats are reserved at invitation time, so a department knows its committed spend before anyone accepts.

Built and working today: organisations, courses, roles, bulk invitation, seat accounting, cohort progress views.

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10 · Why it is defensible

Most AI review tools are a prompt. This is a system.

- **Three-stage pipeline per turn** — private reasoning, visible response,

validation — so a reviewer works out what is **uniquely theirs** to say before saying it

- **Preparation before speaking** — a private memo per reviewer, with the outside literature they consulted

- **Cryptographic assurance** — the certificate id **is** a hash of scores and

ordered evidence

- **Provenance** — which chunk a claim rests on, sha256 re-verified

Scale: 121 API operations · 39 tables · 440 backend tests · ~53,000 lines.

The moat is not the prompt. It is the eleven rules and the measurement harness — each one written after observing a specific failure in real output, not anticipated in advance.

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11 · What does not work yet

Do not skip this slide. It is why the rest is credible.

- **Page citations are not verified** — 391 of 398 chunks in a real database

carry no page number; checking pages would reject legitimate citations

- **Differentiation still degrades by round two** on short papers with few

distinct flaws. The gates catch it; the fix is partial

- **Prep packs cite 3 of 5 sources**, not all five — forcing all five in

produces padding rather than argument

We publish these in the product documentation, with numbers. A tool that tells you it is uncertain is worth more than one that is confidently wrong.

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12 · Roadmap

Now — the review panel, integrity enforcement, assessment, certificates, the institution tier.

Next — record document structure at ingest so page and section citations can be verified fully. Reviewer differentiation beyond round two. LMS integration.

Later — longitudinal readiness across a cohort. Reviewer calibration against real examiner outcomes.

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13 · The ask

Fill in for your audience: capital, a pilot department, a design partner.

What we want from a pilot: one supervisor, one cohort, one term. We measure whether students who rehearse with a panel answer their examiners better than students who do not.

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14 · Close

> Every researcher eventually faces a panel that has read their work carefully and is not being kind about it. There is no reason that should be the first time.

Try it: ten sessions across ten fields, each paper carrying real planted flaws — unconcealed allocation, missing controls, thresholds tuned on the reported data, train/test leakage.

Watch whether the panel finds them.